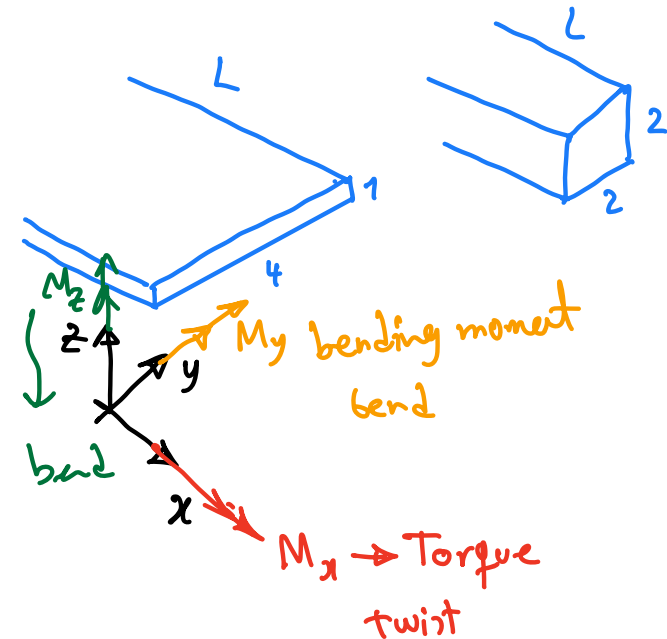


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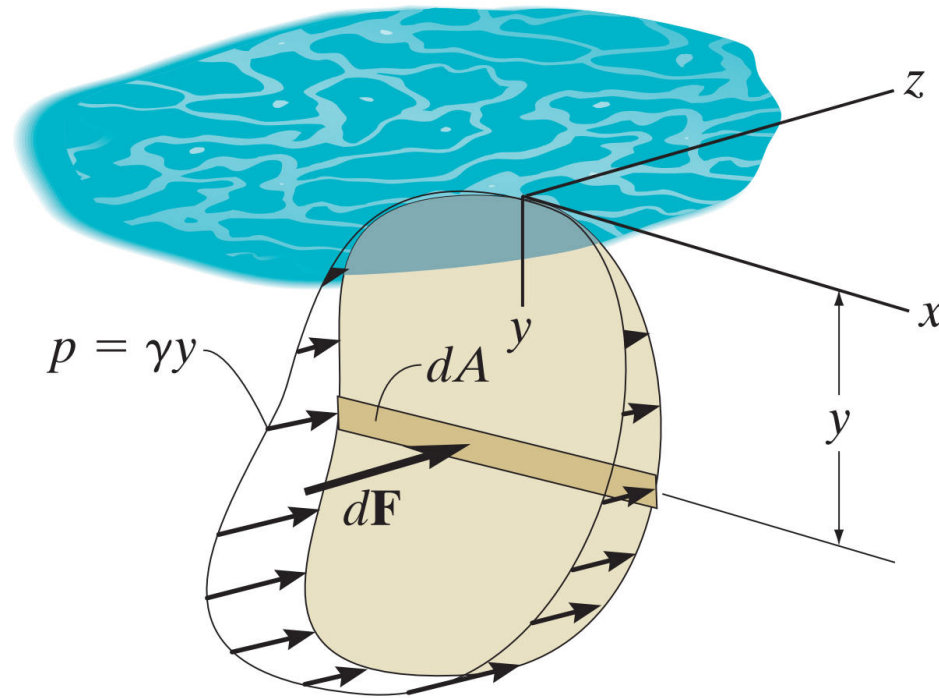
Statics (ME 231)

Lecture 17:

Moments of Inertia

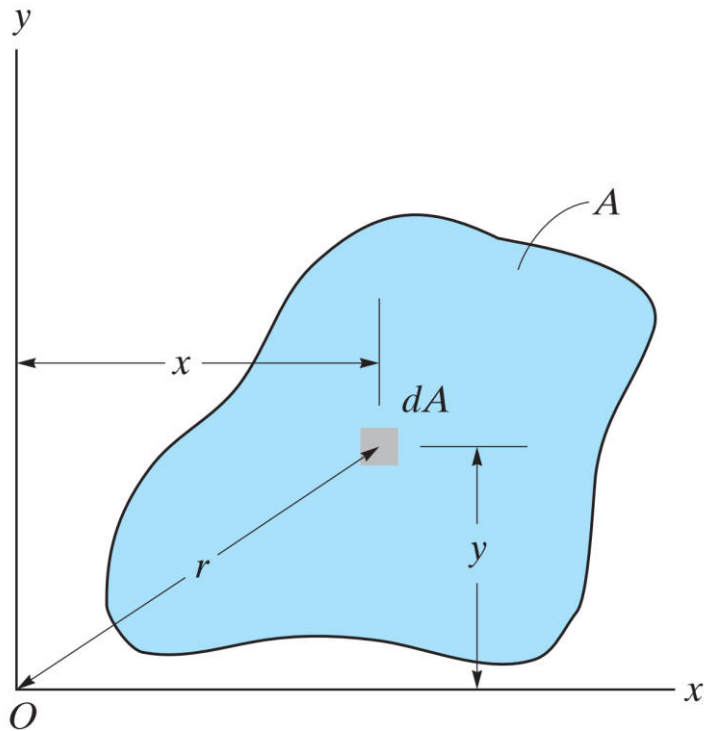
By Saeed Lotfan

Definition of Moments of Inertia for Areas



The integral $\int y^2 dA$ is sometimes referred to as the “**second moment**” of the area about an axis (the x axis), but more often it is called the ***moment of inertia of the area***.

Moment of Inertia of Area

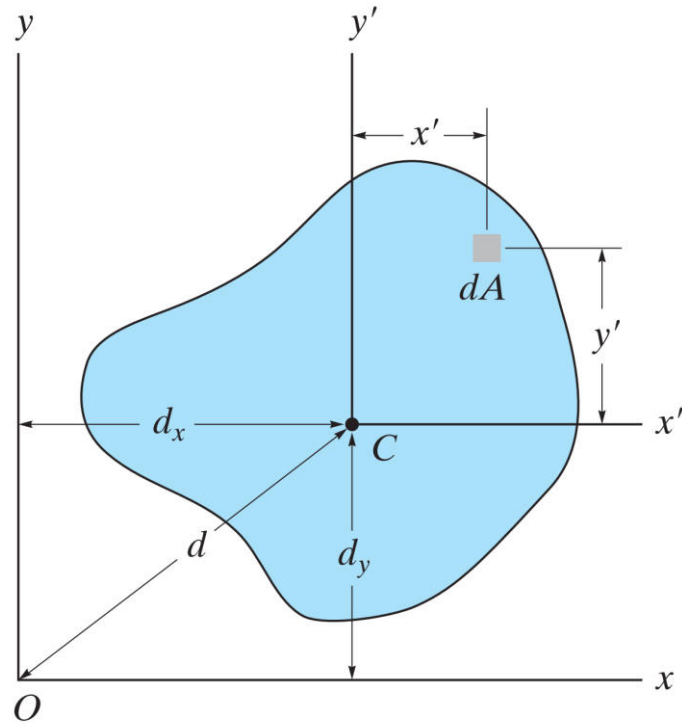


$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

$$J_O = \int_A r^2 dA = I_x + I_y$$

Parallel-Axis Theorem for An Area



$$I_{x'} = \int y'^2 dA, \quad I_{y'} = \int x'^2 dA$$

$$I_x = \int y^2 dA = \int (d_y + y')^2 dA$$

$$I_x = \int_A (y' + d_y)^2 dA \quad \bar{y}' = \frac{\int y' dA}{\int dA}$$

$$= \int_A y'^2 dA + 2d_y \int_A y' dA + d_y^2 \int_A dA$$

$$I_x = I_{x'} + 0 + d_y^2 A$$

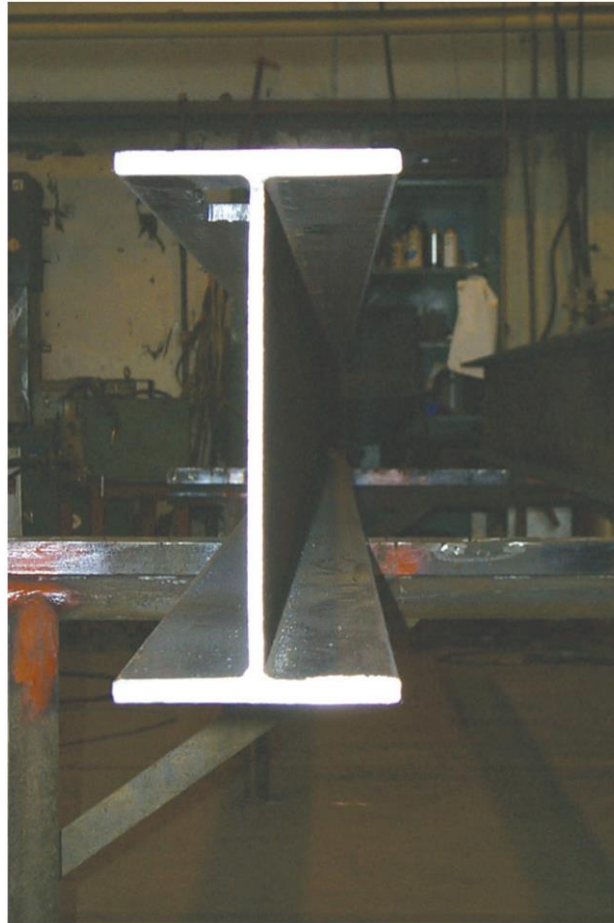
Radius of Gyration of An Area

The *radius of gyration* of an area about an axis has units of length and is a quantity that is often used for the design of columns in structural mechanics.

$$k_x = \sqrt{\frac{I_x}{A}} \quad \rightarrow \quad I_x = k_x^2 A$$

$$k_y = \sqrt{\frac{I_y}{A}}$$

$$k_O = \sqrt{\frac{J_O}{A}}$$



In order to predict the strength and deflection of this beam, it is necessary to calculate the moment of inertia of the beam's cross-sectional area.

Concluding Notes

- The moment of inertia is a geometric property of an area that is used to determine the strength of a structural member or the location of a resultant pressure force acting on a plate submerged in a fluid. It is sometimes referred to as the second moment of the area about an axis, because the distance from the axis to each area element is squared.
- If the moment of inertia of an area is known about its centroidal axis, then the moment of inertia about a corresponding parallel axis can be determined using the parallel-axis theorem.

Example: Determine the moment of inertia for the area shown below with respect to (a) the centroidal x' axis, (b) the axis x_b passing through the base of the rectangle, and (c) the pole or z' axis perpendicular to the $x'y'$ plane and passing through the centroid C.

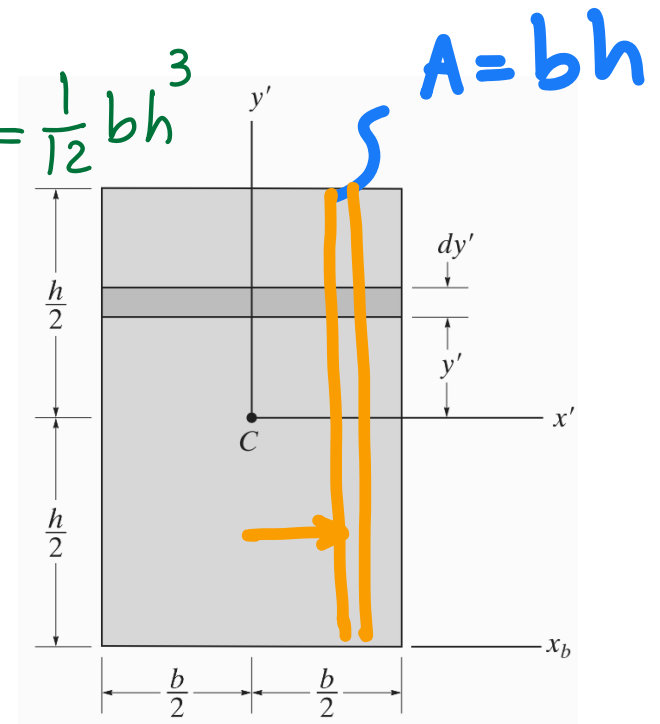
$$I_{x'} = \int dI_{x'} = \int y'^2 dA = \int_{-h/2}^{h/2} y'^2 (b dy') = \frac{1}{12} b h^3$$

$$I_{x_b} = I_{x'} + \left(\frac{h}{2}\right)^2 A = \frac{1}{12} b h^3 + \frac{1}{4} b h^3$$

$$I_{x_b} = \frac{1}{3} b h^3$$

$$I_{y'} = \int x'^2 dA = \int_{-b/2}^{b/2} x'^2 (h dx') = \frac{1}{12} h b^3$$

$$I_{y'} = \int dI_{y'} = \int_{-b/2}^{b/2} \frac{1}{12} dy' b^3 = \frac{1}{12} b^3 h$$



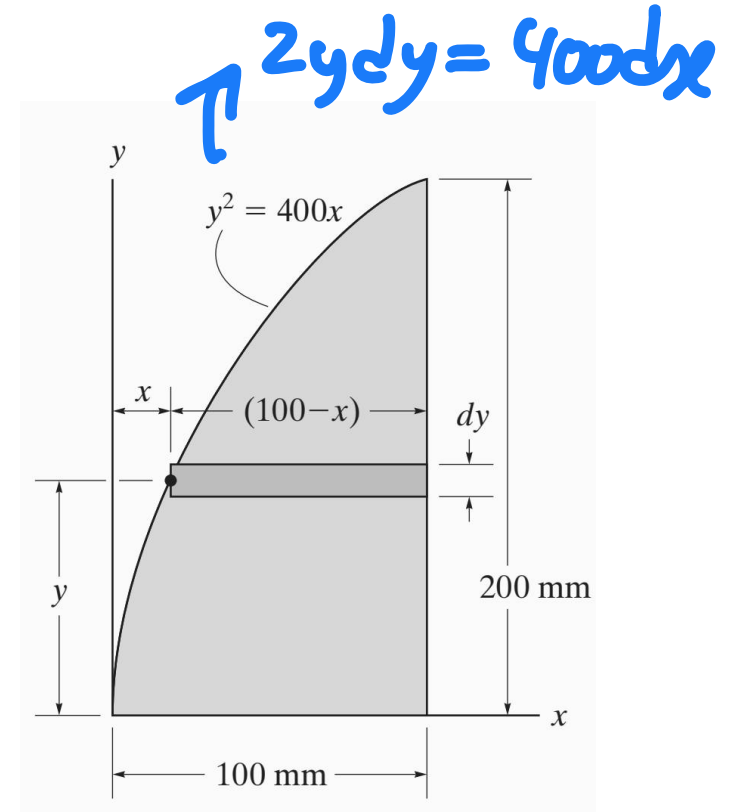
Example: Determine the moment of inertia for the shaded area shown below about the x axis.

$$dI_x = y^2 dA ; dA = (100 - x) dy$$

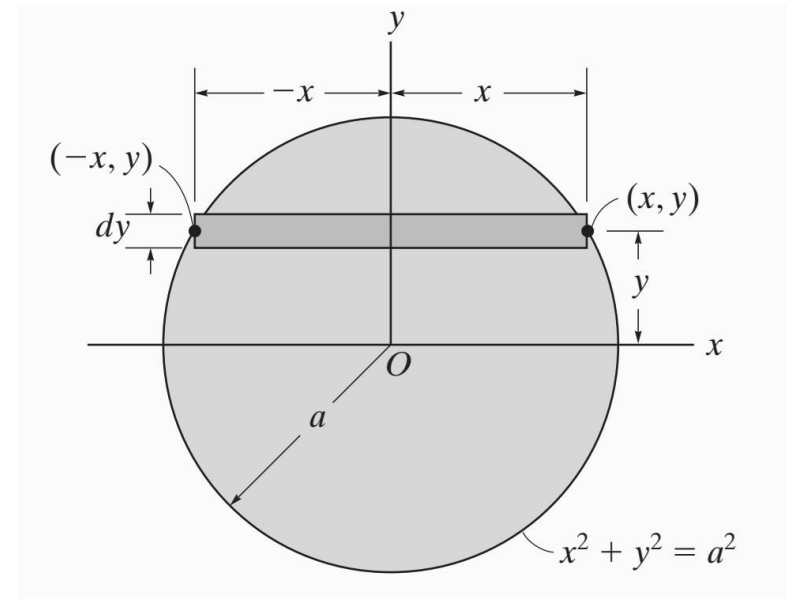
$$I_x = \int y^2 (100 - x) dy$$

$$I_x = \int_0^{100} 400x(100 - x) \frac{200}{20\sqrt{x}} dx$$

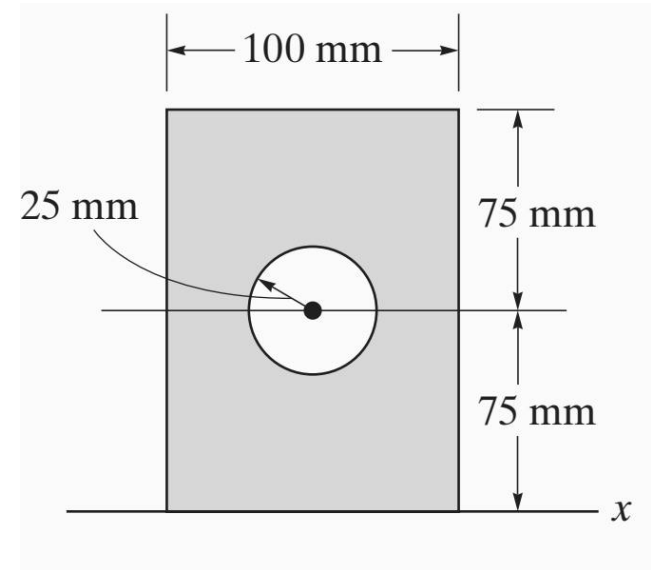
$$\int_0^{100} 4000 x^{\frac{1}{2}} (100 - x) dx$$



Example: Determine the moment of inertia with respect to the x axis for the circular area shown below.

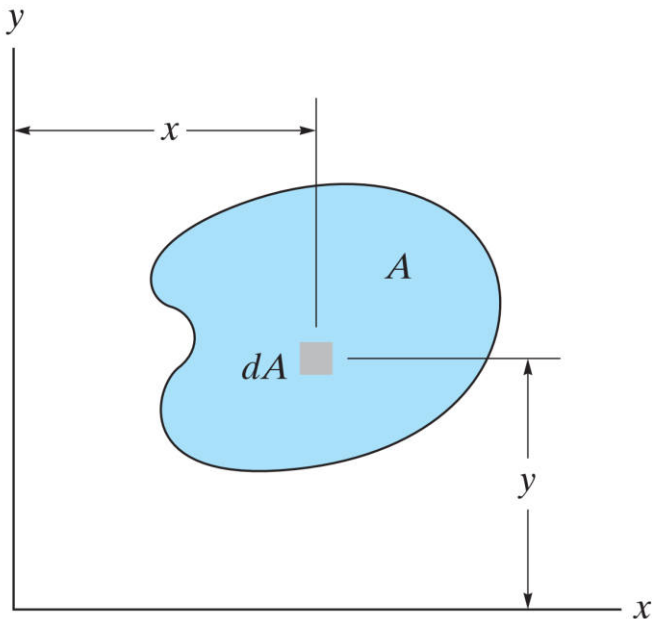


Example: Determine the moment of inertia of the area shown below about the x axis.



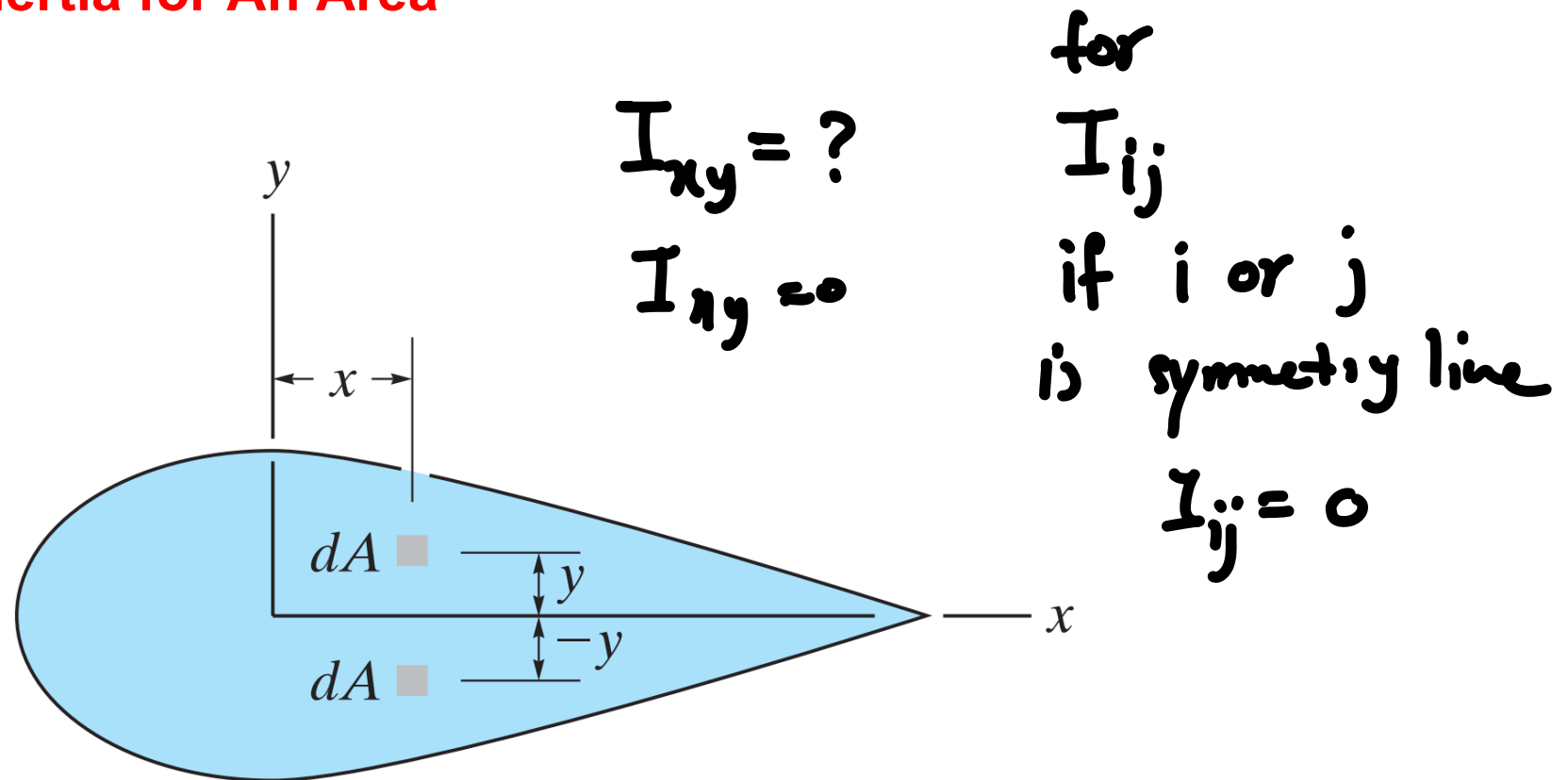
Product of Inertia for An Area

It will be shown in the upcoming lectures that the property of an area, called the product of inertia, is required to determine the *maximum* and *minimum* moments of inertia for the area. These maximum and minimum values are important since they are needed for designing structural and mechanical members such as beams, columns, and shafts.

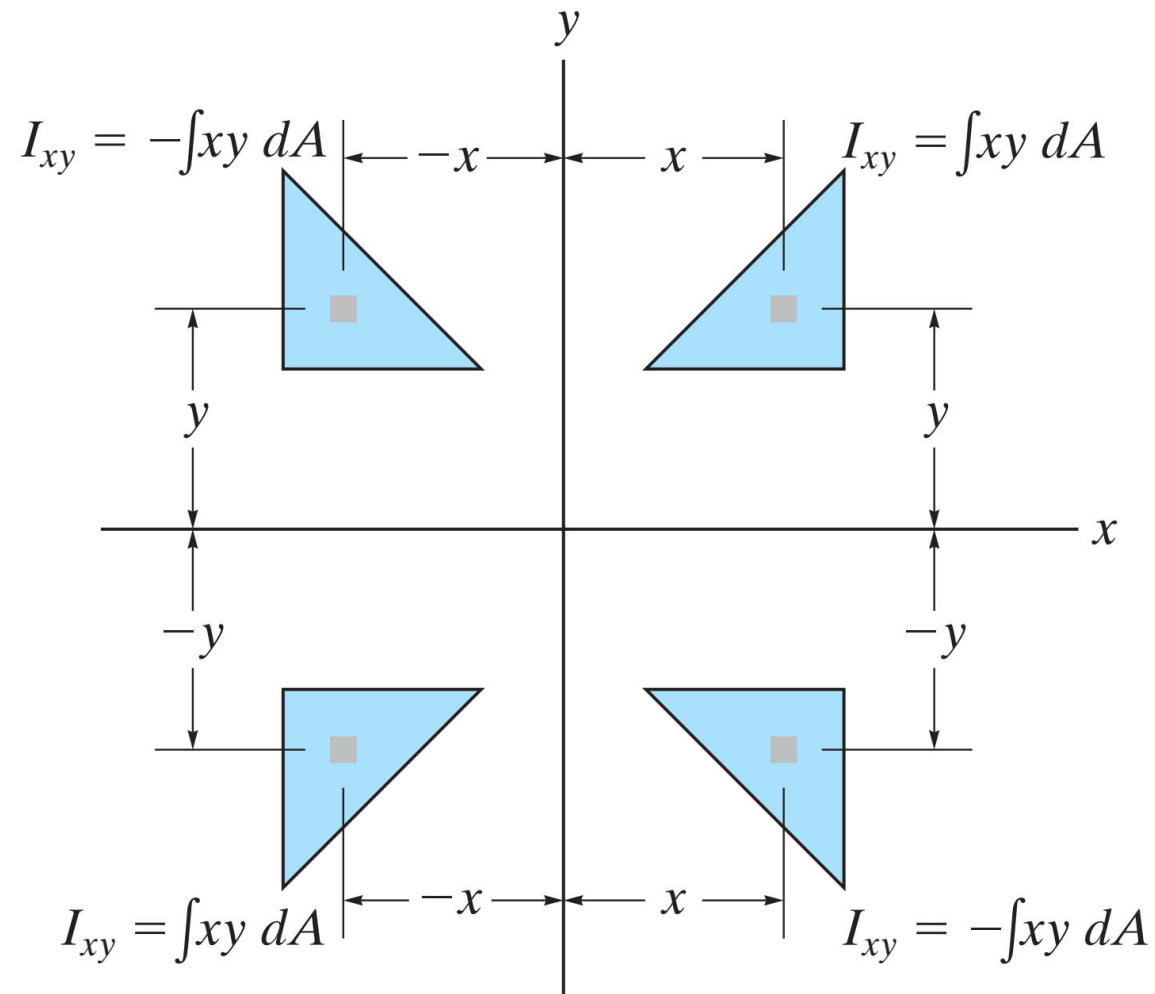


$$I_{xy} = \int_A xy \, dA$$

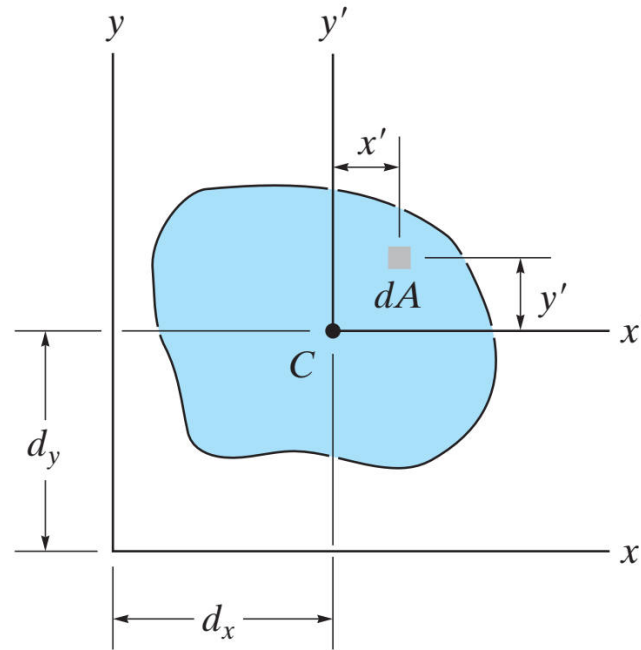
Product of Inertia for An Area



Product of Inertia for An Area



Parallel-Axis Theorem for Product of Inertia for An Area



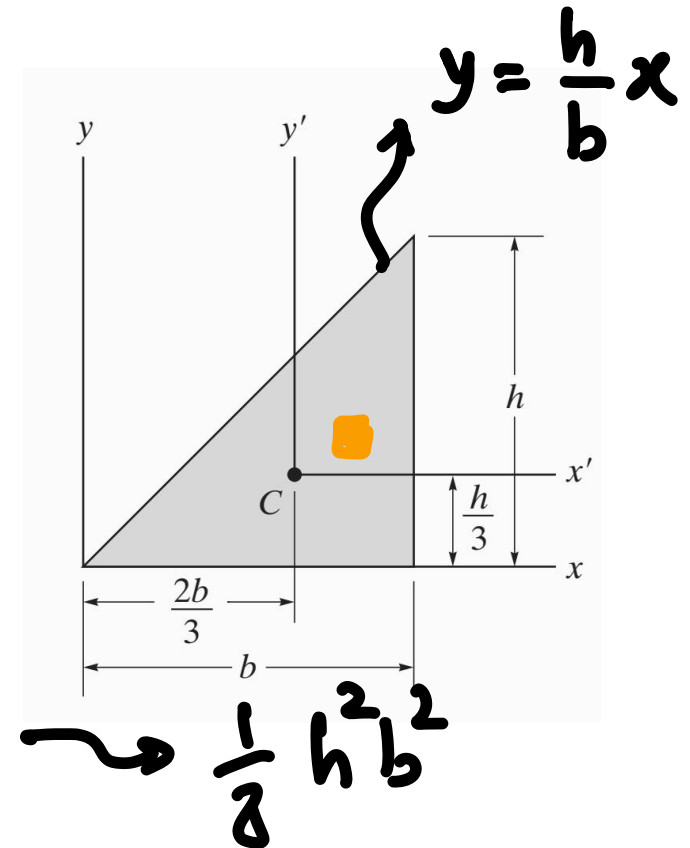
$$\begin{aligned} I_{xy} &= \int_A (x' + d_x)(y' + d_y) dA \\ &= \int_A x'y' dA + d_x \int_A y' dA + d_y \int_A x' dA + d_x d_y \int_A dA \end{aligned}$$

Example: Determine the product of inertia $I_{x'y'}$ for the triangle shown below about axes passing through its centroid C .

$$I_{xy} = I_{x'y'} + \underbrace{\left(\frac{h}{3}\right)}_{dx} \underbrace{\left(\frac{2b}{3}\right)}_{dy} \underbrace{\left(\frac{hb}{2}\right)}_A$$

$$I_{xy} = \int_0^{\frac{h}{b}x} \int_0^b xy \, dx \, dy$$

$$\frac{1}{2} xy^2 \rightarrow \frac{h^2}{2b^2} x^3 \rightarrow \frac{h^2}{8b^2} x^4 \Big|_0^b \rightarrow \frac{1}{8} h^2 b^2$$



Example: Determine the moments of inertia and product of inertia for the cross-sectional area of the member shown below, about the x and y centroidal axes.

$$\begin{aligned}
 I_{xy} &= I_{xy}^{(1)} + I_{xy}^{(2)} + I_{xy}^{(3)} \\
 I_{xy}^{(2)} &= 0 \\
 I_{xy}^{(1)} &= I_{xy_C}^{(1)} + d_x dy A \\
 &= 0 + (-250)(300)A \\
 I_{xy}^{(3)} &= I_{xy_C}^{(3)} + d_x dy A \\
 &= 0 + (250)(-300)A
 \end{aligned}$$

$$I_x = I_{x_1} + I_{x_2} + I_{x_3}$$

$$I_{x_1} = I_{x_C'} + (200)^2(100 \times 300)$$

$$I_{x_1} = \frac{1}{12}(100)(300^3) + (200)^2(100 \times 300)$$

$$I_{x_2} = I_{x_C'} = \frac{1}{12}(600)(100)^3$$

$$I_{x_3} = I_{x_C'} + d^2 A$$

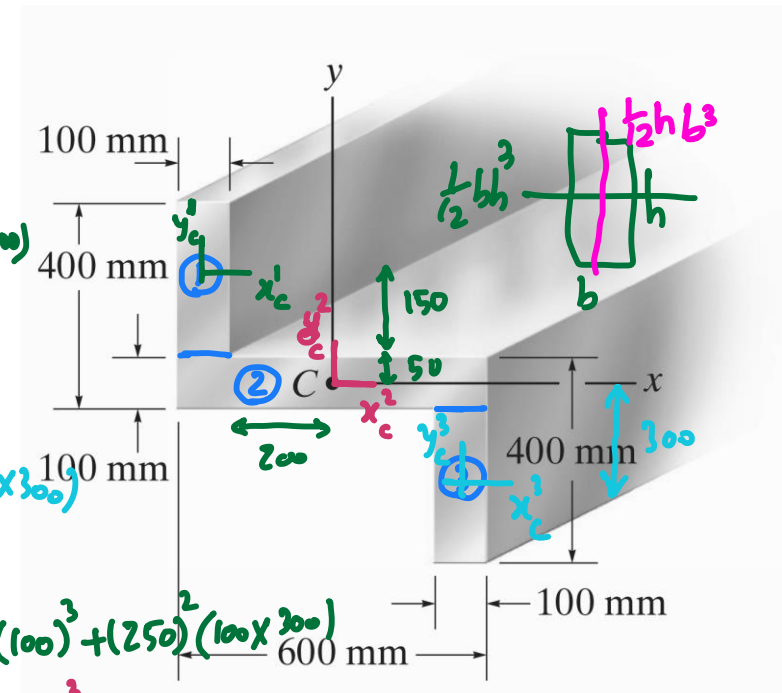
$$= \frac{1}{12}(100)(300^3) + 300^2(100 \times 300)$$

$$I_y = I_{y_1} + I_{y_2} + I_{y_3}$$

$$I_{y_1} = I_{y_C'} + d^2 A = \frac{1}{12}(300)(100^3) + (250)^2(100 \times 300)$$

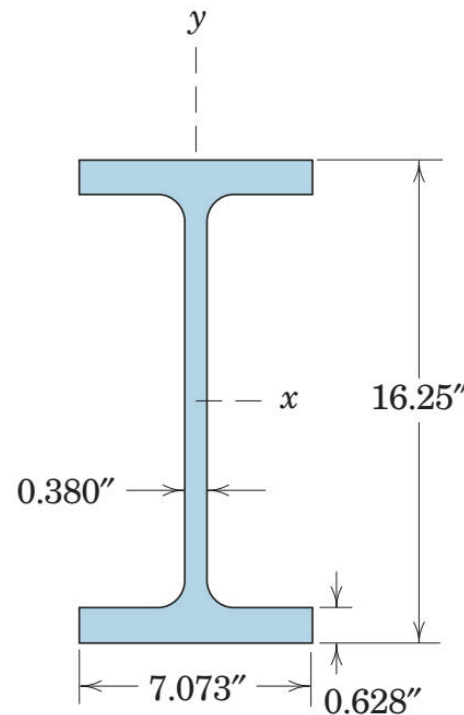
$$I_{y_2} = I_{y_C'} + d^2 A = \frac{1}{12}(100)(600^3) + 0$$

$$I_{y_3} = I_{y_C'} + d^2 A = \frac{1}{12}(300)(100^3) + (250)^2(100 \times 300)$$



Practice at Home

1. The cross-sectional area of a wide-flange I-beam has the dimensions shown. Obtain a close approximation to the handbook value of $I_x = 657 \text{ in.}^4$ by treating the section as being composed of three rectangles.



Practice at Home

2. Determine the polar moment of inertia about point O of the shaded area shown.

